Efficient Randomized Experiments Using Foundation Models

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Problem Setting

- We collect an experimental sample $(X_i, A_i, Y_i)_{i=1}^n$ with fixed treatment probability $\pi = \mathbb{P}(A = 1)$
- We want to estimate $\theta = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0]$
- The average treatment effect is estimated with AIPW

$$\hat{\theta}(\hat{h}) = \sum_{i=1}^{n} \frac{A_i - \pi}{\pi(1-\pi)}(Y_i - \hat{h}(X_i, A_i)) + \hat{h}(X_i, 1) - \hat{h}(X_i, 0)$$

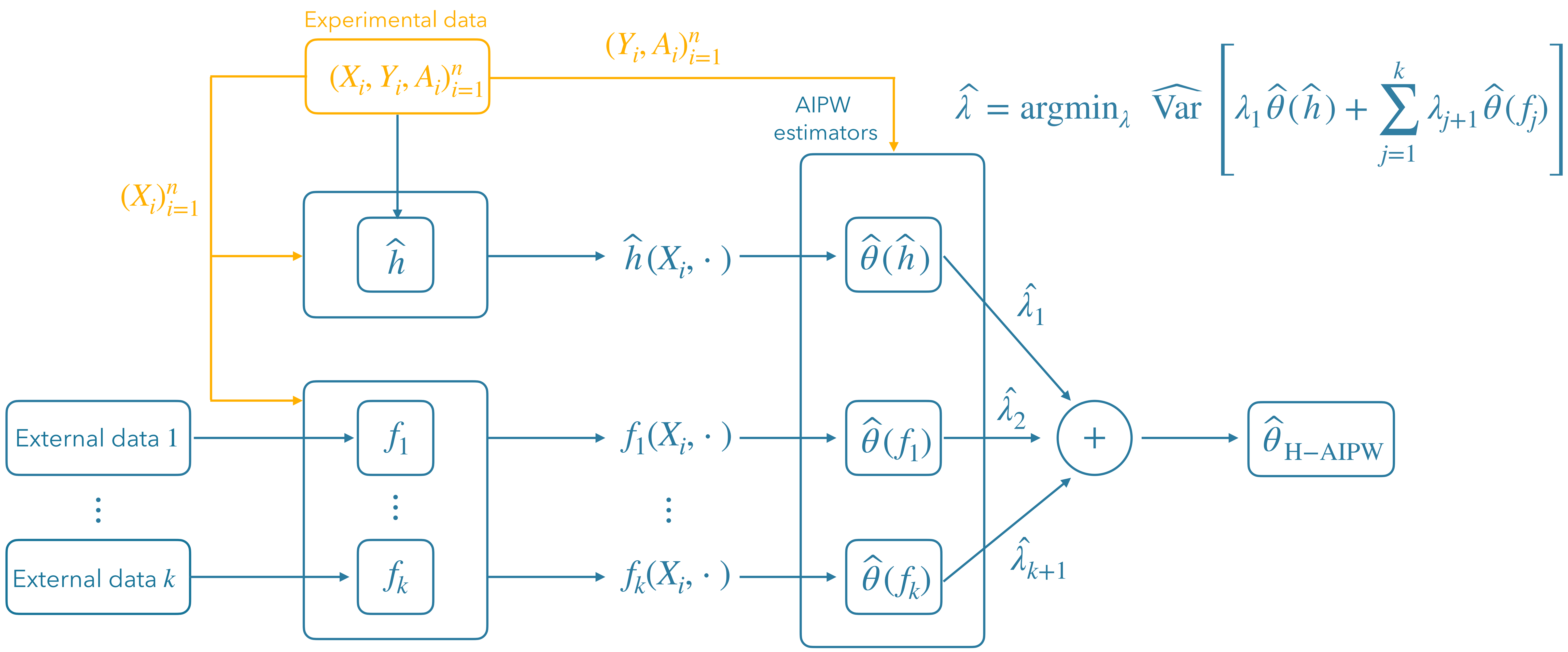
- But the model $\hat{h} = \widehat{\mathbb{E}}[Y \mid X, A]$ is learned from the small experimental sample using e.g. linear regression
  - **Idea:** use external data and flexible models to learn $\hat{h}$

We propose the **H**ybrid **A**ugmented **I**nverse **P**robability **W**eighting **(H-AIPW)**, a novel estimator that safely leverages foundation models to improve efficiency.

☑ Protects against bias from the foundation models

☑ Always at least as efficient as the classic AIPW estimator

Method



Experiments

Estimator	Melin et al. (2022) $n = 100$	Melin et al. (2022) $n = 200$	Silverman et al. (2022) $n = 100$	Silverman et al. (2022) $n = 200$	Kennedy et al. (2020) $n = 100$	Kennedy et al. (2020) $n = 200$	Fahey et al. (2023) $n = 100$	Fahey et al. (2023) $n = 200$
H-AIPW	**10.39**	**10.28**	**2.10**	**2.14**	**17.09**	**17.47**	**4.87**	4.94
PPI-style	11.00	11.06	2.25	2.26	17.87	17.97	4.88	**4.91**
PROCOVA	11.81	10.62	2.24	2.22	18.38	18.11	5.18	5.09
AIPW (boosting)	12.82	12.44	2.82	2.83	23.09	23.12	6.31	6.37
AIPW (standard)	11.72	10.57	2.22	2.20	18.09	17.95	5.09	5.04
DM	11.10	11.10	2.30	2.30	18.07	18.08	5.61	5.62

Example System Prompt

You are a 35-year-old female, politically Democrat, holding liberal views. Additionally, your religion is Christianity, and you once or twice a month attend religious services. You reside in a building with two or more apartments, and your household has a yearly income of $85,000 to $99,999.
You are responding to a scenario reflecting a debate involving college campus events and broader social issues.

Example User Prompt

Treatment: *A student organization denied Antifa's request for a rally, citing safety concerns due to altercations at similar events.*
Outcome question: Do you agree or disagree with the statement:
"Cancel culture is a big problem in today's society." Choose an integer between 1 (strongly agree) and 5 (strongly disagree).
Instruction: Reflect on the scenario and use your reasoning to assign a value.